Akoizo Original Practice - Form 3 - Answer key

Division C | Mission Possible | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) A ramp is 30 cm long and rises 12 cm. What is its IMA?

A. 2.5

B. 0.4

C. 3.6

D. 18

Correct: A

2. (2 points) A device has one dimension of 83 cm. What happens?

A. It runs but is scored in Tier 2, with a 25-point penalty

B. It cannot run

C. Only a 25-point penalty

D. It is scored in Tier 3

Correct: A

3. (2 points) Why does a water timer slow down as it drains?

Unit: Timing: Target Time, the water timer, and pacing.
Worked-case data: Target Time 60 s. The device runs 67 s with a 25-s water timer that ended at 40 s. Calculate time points and timer bonus.
Separate investigation observations:
Timer calibration: 150 mL -> 18 s; 300 mL -> 30 s; 450 mL -> 39 s.
Other actions: Total 46 s without the timer.
Target Time: 80 s announced at State.

A. The hole gets smaller

B. Air pressure rises

C. Outflow speed depends on the water height above the hole

D. Water gets thicker

Correct: C

4. (2 points) Why might Alka-Seltzer timing differ at competition?

Unit: Chemical actions: gas, cold, and fizz.
Worked-case data: You need about 600 mL of CO_2 to inflate a balloon to size. How much baking soda and vinegar should you use?
Separate investigation observations:
Test 1: 1.0 g baking soda + 30 mL vinegar: balloon expanded 14 cm.
Test 2: 2.0 g + 30 mL vinegar: expanded 21 cm.
Test 3: 3.0 g + 30 mL vinegar: expanded 22 cm; undissolved powder left over.

A. The coin order changes it

B. Gravity is different

C. Reaction speed depends on water temperature

D. Tablets are a different color

Correct: C

5. (4 points) Explain why a selective coin trigger can be worth 175 points.

answer: A device that starts only on the quarter earns 200 start points instead of 25, because the supervisor tries both small coins first. (4 points).

6. (4 points) Name the three ASL columns and what each contains.

Unit: The Final Action, the ASL, and competition day.
Worked-case data: Estimate the landing speed of a gum pack dropped 55 cm, and suggest how to keep it on the platform.
Separate investigation observations:
Drop tests: Gum fell 52 cm; landed on platform 7 of 10 times, bouncing off 3.
Change: Added a level release shelf; dropped 51 cm.
New tests: Landed and stayed 10 of 10 times.

answer: Column 1: ordinal number matching the device label; column 2: the object moved; column 3: the rule task number (roman numeral i-xii). (4 points).

7. (4 points) How can a falling magnetic object be released by a previous action?

answer: The previous action slides or lifts the magnet away from the surface, removing the magnetic force so the object falls. (4 points).

8. (4 points) Describe a gravity-driven screw action.

Unit: Simple machines: levers, inclines, pulleys, and screws.
Worked-case data: A screw has a pitch of 1.5 mm. How many turns move an object 2.0 cm, and is that enough rotation for the action?
Separate investigation observations:
Pulley build: Load 420 g lifts 11 cm; effort rope moves 22 cm.
Supervisor check: Counts two supporting segments.
Team claim: Pulley action meets IMA 3.

answer: A falling weight unwinds a string from a spool on a threaded rod, turning it at least two full turns so a non-rotating nut moves an object at least 2 cm. (4 points).

9. (4 points) What happens to scoring if a team touches the device for a fourth time?

answer: Only three touches are allowed; if it stops after the third, it may not be touched again, time is scored as zero, and no further scoring occurs. (4 points).

10. (4 points) What happens to time points if the run reaches twice the Target Time?

answer: The team receives zero time points. (4 points).

11. (6 points) Solve this case and explain your reasoning: You need about 600 mL of CO_2 to inflate a balloon to size. How much baking soda and vinegar should you use?

reasoning: Moles CO_2 = 0.600 L ÷ 24 L/mol = 0.025 mol.
Baking soda: 0.025 mol x 84 g/mol about 2.1 g.
Vinegar: 0.025 mol ÷ 0.83 mol/L about 30 mL; add about 20 percent margin to each. (3 points).

conclusion: About 2.5 g of baking soda with 36 mL of vinegar gives enough gas with margin. (3 points).

12. (2 points) Which gap should the team use?

Supplied synthetic observations:
Gap 21.5 mm: Nickel sometimes stuck on the rail and triggered the start.
Gap 22.8 mm: Dime and nickel fell through 20 of 20; quarter carried 20 of 20.
Gap 24.0 mm: Quarter fell through in 6 of 20 trials.

A. 21.5 mm

B. 24.0 mm

C. 22.8 mm

Correct: C

13. (2 points) Why did the change help?

Supplied synthetic observations:
Drop tests: Gum fell 52 cm; landed on platform 7 of 10 times, bouncing off 3.
Change: Added a level release shelf; dropped 51 cm.
New tests: Landed and stayed 10 of 10 times.

A. The platform got walls

B. The pack fell flat instead of tumbling, so it did not bounce off

C. The drop became much shorter

Correct: B

14. (2 points) Define “IR LED” in the context of Mission Possible.

answer: Light-emitting diode producing infrared light; the required transmitter. (2 points).

15. (2 points) Define “Fulcrum” in the context of Mission Possible.

answer: Pivot point of a lever. (2 points).

16. (2 points) Define “Quarter-only start” in the context of Mission Possible.

answer: Device ignores small coins and starts on the quarter; 200 points. (2 points).

17. (2 points) Define “ASL columns” in the context of Mission Possible.

answer: Ordinal number, object moved, and rule task numeral. (2 points).

18. (2 points) Define “Flow rate” in the context of Mission Possible.

answer: Volume of water passing per second. (2 points).